
CS204: Algorithms & Data Structures Lab

Week # 13 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 04-Nov-2024 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission time: 16:45 Hrs, 04-Nov-2024

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Question 1: (100 points)

Graph Traversal. Write a C++ program for the following tasks. Study the supplementary material file named `graph-representation.pdf` on undirected graph and its representation. Irregular 2D array means, array having different columns in each row. You have to dynamically allocate memory for every row as per number of columns requirement.

Task 01 (10 marks) Declare the following in the header file `graph.h`

1. `namespace cs204_graph` which should include the `class graph`.
2. Declare data members as stated in Task 02
3. Declare all the member functions as stated in Task 02

Task 02 (58 marks) In the file `graph.cc` perform the following:

1. (1 mark) include the header file `graph.h`
2. Define all the members functions of class `graph` with the following specifications
3. (3 marks) Private members
 - (a) (1 mark) An `int` representing number of vertices (n_v)
 - (b) (1 mark) A pointer to pointer to an `int` (for adjacency matrix).
 - (c) (1 mark) A pointer to pointer to an `int` (for adjacency list).
4. (55 marks) Public members
 - (a) (5 marks) A constructor with the following specifications
 - i. **Input argument 1** an `int` representing number of vertices in the graph
 - ii. **Return** None
 - iii. **Functionality** Dynamically create memory for adjacency matrix. Initialize the each element with value 0. Dynamically create memory to hold n_v pointers to `int`. Initialize all of these with value NULL.
 - (b) (2 marks) A function `adjancency_matrix` with the following specifications
 - i. **Input argument 1** Pointer to a `char` denoting the file name
 - ii. **Return** None
 - iii. **Functionality** Reads the data in the specified input file into adjacency matrix data member
 - (c) (4 marks) A function `adjancency_list` with the following specifications

- i. **Input argument 1** Pointer to a `char` denoting the file name
 - ii. **Return** None
 - iii. **Functionality** Reads the data in the specified input file into adjacency list data member
- (d) (10 marks) A function `dfs_matrix` with the following specifications
 - i. **Input arguments** None
 - ii. **Return** None
 - iii. **Functionality** Print (to the standard output) the order of traversal of the given input graph using DFS method and using adjacency matrix
- (e) (10 marks) A function `dfs_list` with the following specifications
 - i. **Input arguments** None
 - ii. **Return** None
 - iii. **Functionality** Print (to the standard output) the order of traversal of the given input graph using DFS method and adjacency list data member
- (f) (10 marks) A function `bfs_matrix` with the following specifications
 - i. **Input arguments** None
 - ii. **Return** None
 - iii. **Functionality** Print (to the standard output) the order of traversal of the given input graph using BFS method and using adjacency matrix data member
- (g) (10 marks) A function `bfs_list` with the following specifications
 - i. **Input arguments** None
 - ii. **Return** None
 - iii. **Functionality** Print (to the standard output) the order of traversal of the given input graph using BFS method and using adjacency list data member
- (h) (4 marks) Destructor with the following specifications
 - i. **Input arguments** None
 - ii. **Return** None
 - iii. **Functionality** Free the memory of the class members

Task 03 (30 marks) Write the `main` function in a file `cs204-graph.cc`. Perform the following

1. (1 mark) Include the graph header file
2. (1 mark) use the `cs204_graph` namespace
3. (1 mark) Create a graph object
4. (5 marks) Infer number of vertices and number of edges from the input file `am-10x10.txt`
5. (1 mark) Perform DFS using adjacency matrix
6. (1 mark) Perform BFS using adjacency matrix

7. (5 marks) Infer number of vertices and number of edges from the input file `al-10x10.txt`
8. (1 mark) Perform DFS using adjacency list
9. (1 mark) Perform BFS using adjacency list
10. (1 mark) Create another graph object
11. (5 marks) Infer number of vertices and number of edges from the input file `am-20x20.txt`
12. (1 mark) Perform DFS using adjacency matrix
13. (1 mark) Perform BFS using adjacency matrix
14. (5 marks) Infer number of vertices and number of edges from the input file `al-20x20.txt`
15. (1 mark) Perform DFS using adjacency list
16. (1 mark) Perform BFS using adjacency list